Wednesday Warm-Up: Unit 5, Electromagnetism

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1 Memory Bank

- $c = f\lambda$ ... The relationship between speed of light, c, the frequency, f, and the wavelength, λ.
- $c = 299,792,458$ m s^{-1}, or $c \approx 0.3$ m ns^{-1}.
- $v = c/n$ ... The actual speed of light, v in a material with index of refraction n.
- $\bar{S} = E^2/(2c\mu_0)$ ... Average **intensity**, or W m^{-2}, of electromagnetic energy.
- $\cos(x)\cos(y) = (1/2)(\cos(x+y)+\cos(x-y))$ ... Trigonometric identity for *mixing* sinusoids.

2 Electromagnetic Waves and Energy

1. (a) What is the wavelength of a 1 GHz radio wave? (b) What is the frequency of a 900 MHz cell phone signal? (c) What is the speed of light in ice, if we observe a frequency of 100 MHz, and a wavelength of 1.69 m?
2. In part (c) of the previous exercise, what is the ratio of the speed of light in ice to the speed found in the memory bank? *This is known as the index of refraction.*
3. Suppose a plane is flying 500 meters above an ice shelf in Antarctica. A radar pulse is fired directly downwards at the ice. The signal enteres the ice, travels for another 500 meters, reflects from the ocean, and returns to the plane. How long before the signal returns? The index of refraction for RF (radio-frequency) waves in ice is 1.78, but it is 1.0 in air.
4. What is the intensity of a wave created by an E-field with magnitude 1000 V m^{-1} at the source?
5. How do you think the previous result scales with distance from the source?

3 Signal Mixing and AM Radio

1. (a) Suppose $f(t) = 4\cos(2\pi f_1 t)$ and $g(t) = 2\cos(2\pi f_2 t)$. Using a formula from the Memory Bank, calculate the frequencies present in the product $f(t)g(t)$. (b) Suppose $f_1 = 1440$ kHz, and $f_2 = 560$ Hz. What frequencies exist in $f(t)g(t)$? (c) Suppose an RLC resonator is built to take an audio signal at f_2 and mix it with a *carrier frequency* at f_1. The product is the *radiated* across space. Graph this radiated signal $f(t)g(t)$ below versus time.